

Forensic accounting and audit quality: Assessing their joint effect on fraud detection in manufacturing firms

By

Mojisola Oladunni Jacob-Udeme & Bukola Opemipo Adesuyi

Department of Management Science, Faculty of Arts, Social and Management Sciences,
Dominion University, Ibadan

Correspondences:

mojio.udeme@gmail.com

adesuyibukola4@gmail.com

Abstract

Corporate fraud continues to undermine the financial integrity, operational efficiency, and sustainability of manufacturing firms, underscoring the need for more effective fraud-detection mechanisms. This study examined the individual and joint effects of forensic accounting and audit quality on fraud detection in manufacturing firms. A descriptive survey research design was adopted, and primary data were collected through a structured questionnaire administered to employees engaged in accounting, auditing, finance, compliance, and management functions within selected manufacturing firms. The research instrument was validated by experts and demonstrated satisfactory reliability before the main survey. Data was analysed using descriptive statistics and Ordinary Least Squares multiple regression analysis. The findings revealed that forensic accounting had a significant positive effect on fraud detection ($\beta = 0.459$, $p = .001$), while audit quality also had a significant positive effect ($\beta = 0.315$, $p = .022$). Furthermore, the regression model indicated that forensic accounting and audit quality jointly explained 48.9% of the variation in fraud detection ($R^2 = 0.489$), with the overall model remaining statistically significant ($F = 22.528$, $p < .001$). The study concludes that integrating forensic accounting with high-quality auditing provides a stronger fraud-detection framework than relying on either mechanism alone. It recommends strengthening forensic accounting units, enhancing auditor independence, adopting digital forensic technologies, and integrating forensic accounting procedures into routine audit engagements to improve organisational accountability and financial integrity in manufacturing firms.

Keywords: Forensic Accounting, Audit Quality, Fraud Detection, Manufacturing Firms, Corporate Fraud, Financial Accountability

Introduction

Corporate fraud remains one of the most significant threats to organisational sustainability, financial integrity, and investor confidence in today's business environment. The increasing digitalisation of business processes, the expansion of global markets, and the growing complexity of corporate operations have created opportunities for sophisticated financial crimes that are often difficult to detect with conventional control mechanisms. Fraudulent activities such as financial statement manipulation, asset misappropriation, bribery, and corruption impose substantial financial losses on organisations while weakening stakeholder confidence and economic growth (Achmad et al., 2024; Sudarmadi, 2023). Manufacturing firms are particularly susceptible because their operations involve extensive procurement systems, complex inventory management, multiple production processes, and high transaction volumes that create opportunities for fraudulent practices to remain concealed (Osunwole, 2020; Su'un & Abduh, 2025).

Fraud in manufacturing firms commonly occurs through inventory manipulation, procurement fraud, payroll irregularities, fictitious suppliers, production cost distortions, and financial statement misrepresentation. These practices reduce profitability, undermine operational efficiency, and damage organisational reputation while discouraging investment and public trust in financial reporting (Prasetiyo et al., 2023; Osunwole, 2020). As business operations become increasingly technology-driven, fraud schemes have evolved in complexity, demanding more sophisticated investigative approaches than traditional auditing can provide.

This growing complexity has elevated the importance of forensic accounting as a specialised discipline for fraud detection and prevention. Unlike conventional auditing, forensic accounting integrates accounting, auditing, investigative techniques, legal knowledge, and digital analytical tools to identify, investigate, and document financial irregularities suitable for litigation and regulatory action (Afriyie et al., 2022). Contemporary forensic accounting extends beyond post-fraud investigations by incorporating proactive fraud risk assessment, continuous monitoring, and digital forensic analysis. Empirical evidence demonstrates that forensic accounting significantly improves fraud detection and strengthens internal control systems within manufacturing organisations (Lawal, 2020; Naz & Khan, 2024).

Audit quality also plays a critical role in promoting financial transparency and accountability. High-quality audits enhance the credibility of financial statements by ensuring auditor independence, professional scepticism, and compliance with recognised auditing standards. Organisations benefiting from high-quality audits generally experience lower levels of financial statement fraud because competent auditors are better positioned to identify material misstatements and irregularities (Almatarneh et al., 2026; Nejad et al., 2024). Moreover, auditors' understanding of forensic accounting principles has been shown to improve audit quality by strengthening fraud risk assessment and investigative capabilities (Janrosl et al., 2025).

Despite improvements in auditing standards, traditional auditing has been criticised for its limited ability to detect sophisticated financial crimes. Conventional audits primarily focus on expressing opinions on financial statements rather than conducting detailed fraud investigations,

allowing complex fraud schemes to remain undetected until significant losses occur (Sudarmadi, 2023; Lawal, 2020). Fraudsters increasingly exploit weaknesses in governance structures, internal controls, and routine audit procedures while using advanced technologies to conceal illicit activities (Anyadufu & Uchechi, 2023; Sabale & Jogdand, 2025).

Although numerous studies have examined forensic accounting and audit quality independently, evidence regarding their combined influence on fraud detection remains limited. Existing research generally confirms that forensic accounting enhances fraud investigations, while audit quality improves the reliability of financial reporting and reduces fraudulent practices (Achmad et al., 2024; Kehinde et al., 2023). However, few empirical studies have examined whether integrating forensic accounting with high-quality auditing yields superior fraud-detection outcomes in manufacturing firms.

Research Objectives

The study seeks to:

- i. Examine the effect of forensic accounting on fraud detection.
- ii. Determine the effect of audit quality on fraud detection.
- iii. Assess the joint effect of forensic accounting and audit quality on fraud detection.

Hypotheses

- i. There is no significant relationship between forensic accounting and fraud detection.
- ii. There is no significant relationship between audit quality and fraud detection.
- iii. There is no significant joint effect of forensic accounting and audit quality on fraud detection.

Fraud Detection

Fraud detection is the systematic process of identifying intentional acts of deception, concealment, or manipulation intended to obtain unlawful financial or non-financial benefits at the expense of an organisation. It involves recognising suspicious transactions, analysing unusual patterns, and collecting evidence to determine whether fraudulent activities have occurred (Rashid et al., 2022; Bhattacharya, 2020). The principal objectives of fraud detection are to promptly identify irregularities, minimise financial losses, strengthen internal controls, support legal proceedings, and discourage future misconduct. Effective fraud detection also enables organisations to address control deficiencies and improve corporate governance by preventing the recurrence of fraudulent practices (Achmad et al., 2024; Petrushka, 2025).

Its importance extends beyond financial recovery, as timely detection protects organisational reputation, enhances stakeholder confidence, and promotes the credibility of financial reporting (Sudarmadi, 2023). Common indicators of fraud include unexplained accounting adjustments, unusual transaction patterns, weak internal controls, poor governance structures, financial distress, and discrepancies between accounting records and physical assets, all of which require further investigation (Huang et al., 2016; Afriyie et al., 2022).

Corporate Fraud in Manufacturing Firms (157 words)

Corporate fraud in manufacturing firms encompasses deliberate acts of deception committed to obtain unlawful financial benefits through production, procurement, inventory management, payroll administration, or financial reporting activities. The operational complexity of manufacturing organisations creates multiple opportunities for fraudulent practices because of extensive supply chains, large inventories, and high transaction volumes (Rashid et al., 2022; Sudarmadi, 2023).

Inventory fraud commonly involves stock theft, diversion, or manipulation of inventory records, while procurement fraud includes supplier collusion, inflated invoices, bribery, and unauthorised purchasing arrangements (Glodstein, 2015; Enofe & Fasua, 2023). Payroll fraud occurs through ghost employees, unauthorised payments, or falsified wage claims. Financial statement fraud involves the intentional manipulation of accounting records to misrepresent organisational performance, whereas asset misappropriation refers to the theft or misuse of cash, inventory, equipment, or other organisational resources (Setiawan, 2024). These fraudulent activities weaken operational efficiency, reduce profitability, distort financial reporting, and diminish stakeholder confidence, making effective fraud detection indispensable within manufacturing firms.

Forensic Accounting (170 words)

Forensic accounting is a specialised field of accounting that integrates accounting, auditing, investigative procedures, and legal knowledge to detect, investigate, prevent, and document financial fraud for possible judicial proceedings. Unlike conventional auditing, forensic accounting focuses on uncovering intentional financial misconduct and producing evidence that is admissible in legal and regulatory investigations (Sudarmadi, 2023; Guellim et al., 2024). The discipline evolved in response to increasingly sophisticated corporate fraud schemes and the limitations of traditional audit procedures in identifying concealed financial crimes.

Advances in technology, digital transactions, and complex business structures have further expanded its scope beyond fraud investigation to include fraud prevention, litigation support, and proactive risk management (Ragini, 2022; Daraojimba et al., 2023). The primary objectives of forensic accounting are to investigate suspected fraud, strengthen internal control systems, support dispute resolution, and enhance organisational accountability (Afriyie et al., 2022). Its distinguishing characteristics include professional scepticism, analytical reasoning, evidence-based investigation, multidisciplinary collaboration, and the ability to combine financial expertise with legal and technological knowledge to resolve complex financial disputes.

Forensic Accounting Techniques

Modern fraud investigations rely on a range of specialised techniques designed to uncover financial irregularities that may escape conventional audit procedures. These techniques combine accounting expertise with investigative, analytical, and technological approaches to generate reliable

evidence for organisational decision-making and legal proceedings (Sudarmadi, 2023; Guellim et al., 2024). Among the most widely used methods is data mining, which analyses large volumes of financial data to identify abnormal transaction patterns and relationships that may indicate fraudulent activity. Through statistical modelling and predictive algorithms, hidden anomalies can be detected more efficiently (Ali et al., 2024).

Digital forensic analysis complements this process by recovering, preserving, and examining electronic evidence from databases, emails, and computer systems, thereby supporting investigations involving technology-enabled fraud (Daraojimba et al., 2023). Another widely recognised technique is Benford's Law, which evaluates the frequency distribution of numerical data to identify potential manipulation or fabricated transactions (Ragini, 2022). Continuous auditing facilitates ongoing monitoring of financial transactions, enabling organisations to detect suspicious activities before significant losses occur (Achmad et al., 2024). Additional methods include investigative interviews, which obtain corroborative evidence from relevant individuals, and fraud risk assessment, which systematically identifies organisational vulnerabilities and recommends preventive control measures to reduce future fraud exposure (Afriyie et al., 2022).

Audit Quality

Effective financial reporting depends largely on the quality of audit processes implemented within an organisation. Audit quality refers to the extent to which audit engagements are conducted in accordance with recognised professional standards while providing reliable assurance on the fairness of financial statements (Sudarmadi, 2023; Dkhar et al., 2025). Auditor independence remains a critical dimension because it enables objective judgment in evaluating financial information without inappropriate managerial influence. Professional competence is equally essential, requiring auditors to possess adequate technical knowledge, practical experience, and continuous professional development to address increasingly complex financial environments (Guellim et al., 2024).

Another important dimension is professional scepticism, which encourages critical evaluation of audit evidence rather than unquestioning acceptance of management representations. Thorough audit planning ensures that high-risk areas receive sufficient attention throughout the engagement, whereas appropriate audit evidence provides a sound basis for audit conclusions and professional opinions (Enofe & Fasua, 2023). Clear audit reporting further enhances transparency by communicating significant findings objectively to stakeholders. Compliance with established auditing standards promotes consistency, ethical conduct, and accountability throughout the audit process. Collectively, these dimensions strengthen confidence in financial reporting and improve organisations' capacity to identify, report, and respond effectively to fraudulent activities before substantial corporate losses occur.

Relationship between Forensic Accounting and Audit Quality

Effective fraud management requires integrating investigative expertise with high-quality audit practices. Although conventional auditing provides independent assurance regarding the

fairness of financial statements, investigative procedures extend beyond routine examinations by identifying suspected fraud and generating evidence suitable for regulatory or legal action (Dkhar et al., 2025; Enofe & Fasua, 2023). Complementary application of both approaches strengthens fraud detection through joint evaluation of suspicious transactions, assessment of internal controls, analytical review of financial information, and comprehensive reporting of irregularities. Greater collaboration between auditors and forensic accounting specialists also enhances the identification of fraud indicators, improves investigative outcomes, and supports stronger governance practices. Such integration ultimately provides organisations with a more robust and sustainable framework for detecting, preventing, and responding to corporate fraud.

Fraud Triangle Theory

Fraud Triangle Theory was developed by Donald Cressey to explain the conditions that motivate individuals to engage in fraudulent behaviour. The theory proposes that fraud is most likely to occur when three elements coexist: perceived pressure, opportunity, and rationalisation. Pressure refers to financial or personal challenges that motivate unethical behaviour, opportunity arises from weaknesses in internal controls or governance systems, while rationalisation enables individuals to justify dishonest actions without perceiving themselves as unethical. The theory has become one of the most influential frameworks for understanding occupational and corporate fraud because it provides a systematic explanation of the behavioural drivers underlying fraudulent activities (Tickner & Button, 2021; Akkeren, 2018). Empirical evidence further demonstrates that pressure and opportunity consistently predict fraudulent practices, although rationalisation remains more difficult to measure directly (Homer, 2019; Skousen et al., 2008).

Within manufacturing firms, the theory explains how financial pressures, operational targets, inventory complexity, and weak internal controls create conditions that encourage fraudulent practices. Large inventories, decentralised procurement systems, and extensive production activities increase opportunities for employees or managers to manipulate records or misappropriate assets. Forensic accounting techniques reduce these opportunities by strengthening fraud investigations, while high-quality auditing enhances monitoring and compliance with established control procedures (Waromi et al., 2025).

The principal strength of Fraud Triangle Theory lies in its ability to provide a practical and widely accepted framework for identifying fraud risk factors across different organisational contexts. Its concepts have been incorporated into auditing standards, fraud risk assessments, and corporate governance practices because they facilitate proactive identification of vulnerable areas (Suryandari et al., 2023). Despite its usefulness, the theory has limitations. It does not adequately explain complex organisational fraud involving collusion, technological manipulation, or institutional failures, and the rationalisation component remains difficult to observe objectively (Schuchter & Levi, 2016).

Agency Theory

Agency Theory explains the relationship between principals, who own organisational resources, and agents, who manage those resources on their behalf. The theory argues that conflicts arise because managers may pursue personal interests that differ from those of shareholders, particularly where information asymmetry limits effective oversight. Such conditions create incentives for earnings manipulation, asset misappropriation, and other fraudulent activities unless appropriate monitoring mechanisms are in place (Jensen & Meckling, 1976; Payne & Petrenko, 2019). Consequently, independent auditing, sound corporate governance, and effective internal controls are considered essential mechanisms for mitigating agency conflicts and promoting accountability (Ajaegbu & Mmayie, 2025; Sunday, 2026).

The theory is highly applicable to manufacturing firms where managers exercise considerable discretion over procurement, inventory management, production costs, and financial reporting. High-quality audits and forensic accounting serve as complementary monitoring mechanisms that reduce information asymmetry, strengthen transparency, and improve the credibility of financial information. Through rigorous examination of financial transactions and internal controls, these mechanisms discourage opportunistic behaviour and enhance fraud detection (Neiroukh & Çaglar, 2025).

A major strength of Agency Theory is its clear explanation of the governance role of auditing in protecting shareholders' interests while supporting accountability within organisations (Wardoyo et al., 2021). Nevertheless, the theory has been criticised for assuming that individuals are primarily motivated by self-interest and for placing excessive emphasis on shareholder interests while giving limited attention to broader stakeholder relationships and ethical considerations (Zogning, 2023).

Theory Adopted and Justification

This study adopts Fraud Triangle Theory as its principal theoretical foundation because it directly explains the behavioural conditions that facilitate fraud within manufacturing firms. The concepts of pressure, opportunity, and rationalisation closely align with the study variables by illustrating how forensic accounting reduces opportunities for fraud through specialised investigation, while audit quality strengthens oversight, professional scepticism, and compliance with auditing standards. Although Agency Theory provides valuable insight into governance relationships and monitoring responsibilities, Fraud Triangle Theory offers a more direct explanation of the causes of fraudulent behaviour and the mechanisms through which it can be detected. Consequently, it provides the most appropriate framework for examining the individual and joint effects of forensic accounting and audit quality on fraud detection.

Empirical Review

Recent empirical evidence indicates that forensic accounting and audit quality contribute to fraud detection, although the magnitude and consistency of their effects vary across institutional and industrial contexts. Most international studies report that forensic accounting strengthens fraud detection by combining investigative expertise, litigation support, digital analytics, and fraud risk

assessment with conventional accounting procedures. Evidence from Pakistan demonstrated that forensic accounting techniques, including fraud investigation and litigation support, significantly improved fraud detection and prevention among business organisations (Naz & Khan, 2024). Similarly, a systematic review concluded that forensic accounting consistently enhances fraud detection despite challenges associated with implementation, technological capacity, and professional awareness (Kaur et al., 2022).

Research on audit quality presents comparatively mixed findings. Evidence from Indonesia showed that higher audit quality reduced financial statement fraud, particularly where organisations engaged reputable auditors and maintained longer audit tenure (Nejad et al., 2024). Comparable findings were reported in Jordan, where higher-quality external audits significantly reduced illegal accounting practices among firms with effective governance structures (Almatarneh et al., 2026). However, another Indonesian study found that audit quality alone did not significantly influence fraud prevention, although internal audit effectiveness and internal control systems produced positive outcomes (Lubis et al., 2024). These inconsistencies suggest that audit quality may be enhanced when supported by complementary governance and fraud-investigation mechanisms.

Evidence from Africa is dominated by Nigerian studies, in which forensic accounting consistently demonstrates positive effects on fraud detection and control. Empirical investigations have shown that forensic auditing significantly strengthens fraud detection within the public sector while improving financial reporting quality and organisational accountability (Oyerogba, 2021; Anthony & Gavine, 2022). Similar findings indicate that forensic accounting enhances fraud management across financial institutions through specialised investigative skills and analytical techniques (Eko, 2022). Within manufacturing firms, studies consistently report that forensic accounting improves fraud detection, reinforces internal controls, and reduces opportunities for financial misconduct (Lawal, 2020; Osunwole, 2020; Anyadufu & Uchechi, 2023; Kehinde et al., 2023).

Recent studies further demonstrate the growing importance of technology-driven forensic accounting. Digital forensic tools, data analytics, and computer-assisted investigations have significantly strengthened fraud detection capabilities within Nigerian manufacturing firms and financial institutions (David et al., 2026). Likewise, technology-oriented forensic accounting practices enhanced fraud prevention and institutional integrity in Nigerian microfinance institutions, while the engagement of forensic accountants substantially increased the likelihood of fraud detection within deposit money banks (Eghe-Ikhurhe et al., 2024; Garba, 2024). Nevertheless, recent reviews observe that most empirical studies continue to rely on questionnaire-based cross-sectional designs, limiting opportunities to establish causal relationships and long-term effectiveness (Jones & Onatuyeh, 2019; Oyedotun et al., 2025).

Despite the growing literature, an important empirical gap remains. Existing studies largely examine forensic accounting and audit quality as separate determinants of fraud detection, with limited evidence regarding their combined influence, particularly within manufacturing firms.

Consequently, this study addresses this gap by examining the joint effect of forensic accounting and audit quality on fraud detection in manufacturing organisations.

Conceptual Framework

The conceptual framework illustrates the hypothesised relationship between the independent variables forensic accounting and audit quality and the dependent variable, fraud detection. The framework assumes that effective forensic accounting practices enhance the identification, investigation, and prevention of fraudulent activities through specialised investigative techniques and analytical procedures. Similarly, high audit quality strengthens fraud detection by promoting auditor independence, professional scepticism, competence, compliance with auditing standards, and sufficient audit evidence. When integrated, both mechanisms are expected to complement one another, producing stronger fraud detection outcomes than either approach operating independently.

Figure 2.1 presents the conceptual relationship underpinning this study.

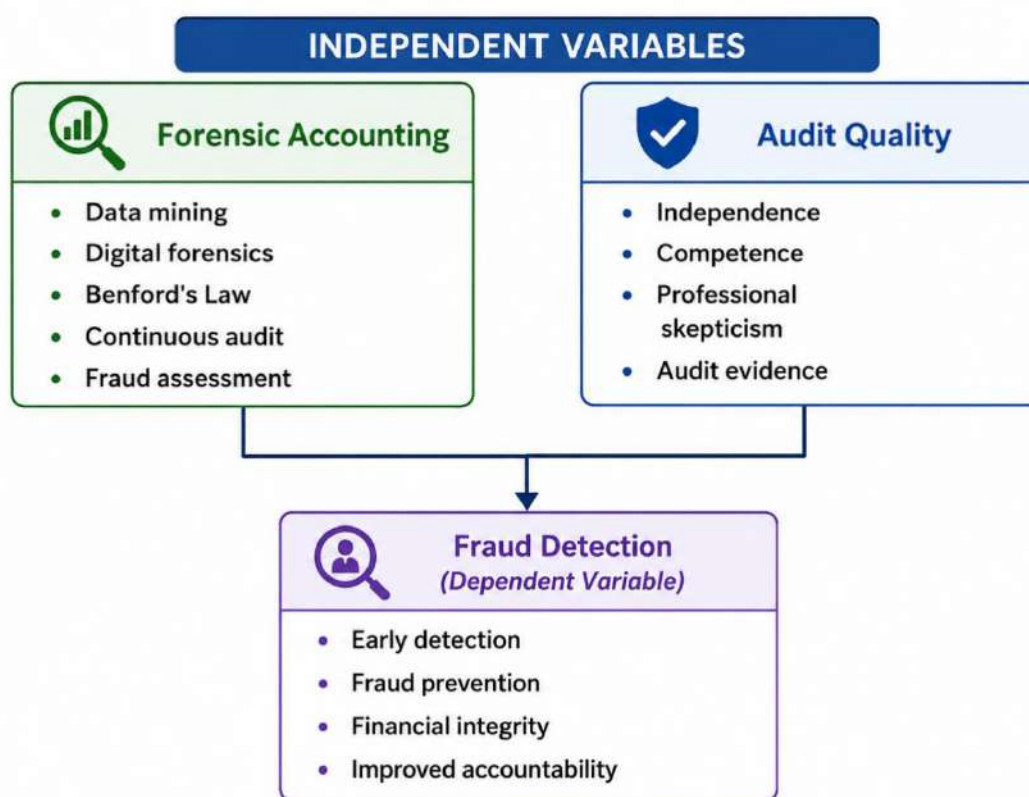


Figure 1: Conceptual Framework Source: Authors' Conceptualisation (2026).

Figure 2.1 illustrates the conceptual relationship proposed in this study. It shows that forensic accounting and the audit quality function as independent variables influencing fraud detection, the

dependent variable. The framework assumes that integrating forensic accounting techniques with high-quality auditing enhances the timely identification, prevention, and investigation of fraud, thereby improving financial integrity and organisational accountability in manufacturing firms.

Methodology

This study adopted a Descriptive Survey Research Design because it provides an appropriate framework for collecting primary data from respondents through structured questionnaires without manipulating the study variables. The design is widely applied in accounting and management research to examine existing organisational practices and determine relationships among variables. It was considered suitable for this study because it enabled the researcher to investigate the effects of forensic accounting and audit quality on fraud detection, drawing on employees' opinions and experiences within manufacturing firms.

The population comprised employees of selected manufacturing firms in Nigeria who were directly involved in accounting, finance, auditing, internal control, compliance, and managerial functions. These categories included accountants, finance officers, internal auditors, external audit liaison officers, compliance officers, and management personnel responsible for financial reporting and corporate governance. The population was considered appropriate because these employees possess adequate knowledge and practical experience regarding forensic accounting practices, audit quality, and fraud detection mechanisms within their organisations.

A representative sample was drawn from the study population using purposive sampling. This non-probability sampling method was adopted because the study required respondents with relevant professional knowledge and practical experience relating to forensic accounting, auditing, and fraud detection. The sample size was considered adequate to yield reliable information for statistical analysis while ensuring that respondents had sufficient expertise to provide informed responses to the variables under investigation.

The study relied exclusively on primary data obtained through a structured questionnaire administered to selected respondents at the participating manufacturing firms. The use of primary data enabled the researcher to obtain first-hand information specific to the study's objectives, thereby enhancing the relevance and reliability of the findings.

Data were collected using a researcher-developed structured questionnaire titled Forensic Accounting, Audit Quality and Fraud Detection Questionnaire (FAAQFDQ). The instrument employed a five-point Likert scale ranging from Strongly Agree (5) to Strongly Disagree (1) and consisted of four sections. Section A obtained respondents' demographic information, including educational qualifications, department, job position, and years of work experience. Section B measured forensic accounting practices using items relating to data mining, digital forensic analysis, Benford's Law, continuous auditing, investigative interviews, and fraud risk assessment. Section C assessed audit quality through auditor independence, professional competence, professional scepticism, audit planning, audit evidence, audit reporting, and compliance with auditing standards.

Section D measured fraud detection using indicators such as early fraud identification, fraud prevention, reporting effectiveness, financial integrity, and organisational accountability.

The instrument's validity was established through Face and Content Validity. Experts in Accounting, Measurement, and Evaluation examined the questionnaire to assess the relevance, clarity, adequacy, and appropriateness of the items in addressing the study objectives. Their observations and recommendations were incorporated into the final version of the instrument.

Reliability was established through a pilot study conducted with respondents who had characteristics similar to those of the study population but were excluded from the main survey. Data obtained from the pilot study were analysed using Cronbach's Alpha with the Statistical Package for the Social Sciences (SPSS). The reliability coefficient exceeded the acceptable benchmark of **0.70**, confirming satisfactory internal consistency and indicating that the instrument was suitable for the study.

Copies of the questionnaire were administered personally to respondents in the selected manufacturing firms. Completed questionnaires were retrieved immediately after completion or collected during subsequent visits to maximise the response rate. Participation was voluntary, informed consent was obtained from all respondents, confidentiality and anonymity were guaranteed, and the information collected was used solely for academic purposes.

Model Specification

The study employed Ordinary Least Squares (OLS) multiple regression analysis to examine the individual and joint effects of forensic accounting and audit quality on fraud detection at the 5% significance level. The model is specified as:

$$[FD = \beta_0 + \beta_1 FA + \beta_2 AQ + \mu]$$

Where:

- a. **FD** = Fraud Detection
- b. **FA** = Forensic Accounting
- c. **AQ** = Audit Quality
- d. **β_0** = Constant term
- e. **β_1 – β_2** = Regression coefficients
- f. **μ** = Error term

Response Rate

A total of 50 questionnaires were administered to respondents drawn from the selected manufacturing firm. All questionnaires were completed and returned, yielding a response rate of 100%. The complete retrieval of the research instrument eliminated the possibility of non-response bias and provided sufficient data for subsequent statistical analyses.

The respondents comprised professionals in accounting, finance, auditing, and management. Female respondents accounted for 52.0% of the sample, while males represented 48.0%. The majority (62.0%) were between 25 and 34 years of age. Approximately 48.0% possessed bachelor's degrees,

whereas 28.0% held professional accounting qualifications. Most respondents had between Predictors: (Constant), Forensic Accounting, Audit Quality

five and ten years of work experience and were employed in accounting, finance, internal audit, and management departments, indicating that they possessed adequate knowledge of forensic accounting practices, audit quality, and fraud detection within manufacturing organisations.

Joint Effect of Forensic Accounting and Audit Quality on Fraud Detection

Multiple regression analysis was conducted to examine the individual and joint effects of forensic accounting and audit quality on fraud detection.

Table 1: Model Summary

Model	R	R Square	Adjusted Square	R- Std. Error of the Estimate
1	0.700	0.489	0.468	0.439

Table 1 presents the regression model summary for the joint effect of forensic accounting and audit quality on fraud detection. The correlation coefficient ($R = 0.700$) indicates a strong positive relationship between the independent variables and fraud detection. The coefficient of determination ($R^2 = 0.489$) indicates that 48.9% of the variation in fraud detection is explained jointly by forensic accounting and audit quality, while the remaining 51.1% is attributable to other factors not included in the model.

Overall Significance of the Regression Model

The significance of the regression model was assessed using analysis of variance (ANOVA). The results are presented in Table 2.

Table 2: Analysis of Variance (ANOVA)

Source	Sum of Squares	Df	Mean Square	F	Sig.
Regression	8.702	2	4.351	22.528	<0.001
Residual	9.078	47	0.193		
Total	17.780	49			

Table 2 presents the results of the ANOVA used to evaluate the overall significance of the regression model. The model produced an F-statistic of 22.528 with a significance level of $p < 0.001$, indicating that the regression model is statistically significant. This finding demonstrates that forensic accounting and audit quality jointly exert a significant influence on fraud detection in manufacturing firms. Consequently, the null hypothesis that the independent variables have no combined effect on fraud detection is rejected, confirming that the model provides a good fit for explaining variations in fraud detection.

Regression Coefficients

Table 3: Regression Coefficients

Variable	Unstandardised Coefficient (B)	Std. Error	Standardised Beta (β)	t-value	Sig.
Constant	0.094	0.879	—	0.107	.915
Forensic Accounting	0.350	0.101	0.459	3.460	.001
Audit Quality	0.587	0.247	0.315	2.373	.022

Dependent Variable: Fraud Detection

Table 3 presents the regression coefficients for the individual effects of forensic accounting and audit quality on fraud detection. The coefficient for forensic accounting ($\beta = 0.459$, $B = 0.350$, $t = 3.460$, $p = .001$) indicates a positive and statistically significant effect on fraud detection, suggesting that greater adoption of forensic accounting practices enhances manufacturing firms' ability to detect fraudulent activities. Audit quality also exhibits a positive and significant influence ($\beta = 0.315$, $B = 0.587$, $t = 2.373$, $p = .022$), implying that improvements in audit quality contribute to more effective fraud detection. Comparatively, forensic accounting demonstrates a stronger predictive effect, as reflected by its higher standardised beta coefficient. The estimated regression equation is expressed as:

$$[FD = 0.094 + 0.350FA + 0.587AQ + \mu]$$

Where:

FD = Fraud Detection

FA = Forensic Accounting

AQ = Audit Quality

μ = Error term.

Summary of Hypotheses Testing

The hypotheses were tested using the regression coefficients and overall model significance. The results are summarised in Table 5.

Table 4: Summary of Hypotheses Testing

Hypothesis	Test Statistic	p-value	Decision
H ₁ : Forensic accounting has no significant effect on fraud detection.	$\beta = 0.459$; $t = 3.460$	.001	Rejected
H ₂ : Audit quality has no significant effect on fraud detection.	$\beta = 0.315$; $t = 2.373$	.022	Rejected
H ₃ : Forensic accounting and audit quality have no joint effect on fraud detection.	$F(2,47) = 22.528$	<.001	Rejected

Source: Authors' Computation (2026).

Table 4, summarises the results of the hypotheses testing. The findings indicate that forensic accounting has a positive and statistically significant effect on fraud detection ($\beta = 0.459$, $t = 3.460$, $p = .001$), leading to the rejection of the first null hypothesis. Similarly, audit quality significantly influences fraud detection ($\beta = 0.315$, $t = 2.373$, $p = .022$), resulting in the rejection of the second null hypothesis. The joint effect of forensic accounting and audit quality is also statistically significant, as indicated by the ANOVA result ($F(2,47) = 22.528$, $p < .001$). Accordingly, the third null hypothesis is rejected, confirming that both variables collectively enhance fraud detection in manufacturing firms.

H₀₁: There is no relationship between Forensic Accounting and Audit Quality Fraud Detection

Multiple regression analysis was conducted to examine the individual and joint effects of forensic accounting and audit quality on fraud detection.

Table 5: Model Summary

Model	R	R Square	Adjusted Square	R-	Std. Error of the Estimate
1	0.700	0.489	0.468		0.439

Predictors: (Constant), Forensic Accounting, Audit Quality

Table 5 presents the regression model summary for the joint effect of forensic accounting and audit quality on fraud detection. The correlation coefficient ($R = 0.700$) indicates a strong positive relationship between the independent variables and fraud detection. The coefficient of determination ($R^2 = 0.489$) indicates that 48.9% of the variation in fraud detection is explained jointly by forensic accounting and audit quality, while the remaining 51.1% is attributable to other factors not included in the model.

Overall Significance of the Regression Model

The significance of the regression model was assessed using analysis of variance (ANOVA). The results are presented in Table 6.

Table 6: Analysis of Variance (ANOVA)

Source	Sum of Squares	Df	Mean Square	F	Sig.
Regression	8.702	2	4.351	22.528	<0.001
Residual	9.078	47	0.193		
Total	17.780	49			

Table 6 presents the results of the ANOVA used to evaluate the overall significance of the regression model. The model produced an F-statistic of 22.528 with a significance level of $p < 0.001$, indicating that the regression model is statistically significant. This finding demonstrates that forensic accounting and audit quality jointly exert a significant influence on fraud detection in manufacturing firms.

Consequently, the null hypothesis that the independent variables have no combined effect on fraud detection is rejected, confirming that the model provides a good fit for explaining variations in fraud detection.

Regression Coefficients

Table 7: Regression Coefficients

Variable	Unstandardised Coefficient (B)	Std. Error	Standardised Beta (β)	t-value	Sig.
Constant	0.094	0.879	—	0.107	.915
Forensic Accounting	0.350	0.101	0.459	3.460	.001
Audit Quality	0.587	0.247	0.315	2.373	.022

Dependent Variable: Fraud Detection

Table 7 presents the regression coefficients for the individual effects of forensic accounting and audit quality on fraud detection. The coefficient for forensic accounting ($\beta = 0.459$, $B = 0.350$, $t = 3.460$, $p = .001$) indicates a positive and statistically significant effect on fraud detection, suggesting that greater adoption of forensic accounting practices enhances manufacturing firms' ability to detect fraudulent activities. Audit quality also exhibits a positive and significant influence ($\beta = 0.315$, $B = 0.587$, $t = 2.373$, $p = .022$), implying that improvements in audit quality contribute to more effective fraud detection.

Comparatively, forensic accounting demonstrates a stronger predictive effect, as reflected by its higher standardised beta coefficient.

The estimated regression equation is expressed as:

$$[FD = 0.094 + 0.350FA + 0.587AQ + \mu]$$

where:

FD = Fraud Detection

FA = Forensic Accounting

AQ = Audit Quality

μ = Error term.

Summary of Hypotheses Testing

The hypotheses were tested using the regression coefficients and overall model significance. The results are summarised in Table 5.

Table 8: Summary of Hypotheses Testing

Hypothesis	Test Statistic	p-value	Decision
H ₁ : Forensic accounting has no significant effect on fraud detection.	$\beta = 0.459$; $t = 3.460$	.001	Rejected

H ₂ : Audit quality has no significant effect on fraud detection.	$\beta = 0.315$; $t = 2.373$	.022	Rejected
H ₃ : Forensic accounting and audit quality have no joint effect on fraud detection.	$F(2,47) = 22.528$	<.001	Rejected

Source: Authors' Computation (2026).

Table 8 summarises the results of the hypotheses testing. The findings indicate that forensic accounting has a positive and statistically significant effect on fraud detection ($\beta = 0.459$, $t = 3.460$, $p = .001$), leading to the rejection of the first null hypothesis. Similarly, audit quality significantly influences fraud detection ($\beta = 0.315$, $t = 2.373$, $p = .022$), resulting in the rejection of the second null hypothesis. The joint effect of forensic accounting and audit quality is also statistically significant, as indicated by the ANOVA result ($F(2,47) = 22.528$, $p < .001$). Accordingly, the third null hypothesis is rejected, confirming that both variables collectively enhance fraud detection in manufacturing firms.

Discussion of Findings

The present study established that forensic accounting exerts a significant positive influence on fraud detection in manufacturing firms, demonstrating that organisations implementing stronger forensic accounting practices are better positioned to identify and investigate fraudulent activities. The finding suggests that specialised investigative procedures strengthen fraud detection by complementing conventional accounting and auditing processes with analytical techniques that uncover concealed financial irregularities. Manufacturing firms typically operate complex procurement systems, maintain extensive inventories, and employ multiple production processes, creating opportunities for financial misconduct. Consequently, integrating investigative expertise into routine financial oversight enhances organisational capacity to identify suspicious transactions before they develop into significant financial losses.

This finding is consistent with the growing body of empirical evidence demonstrating the effectiveness of forensic accounting in strengthening fraud management. Lawal (2020), Osunwale (2020), Anyadufu and Uchechi (2023), Kehinde et al. (2023), and David et al. (2026) all reported significant positive relationships between forensic accounting practices and fraud detection in Nigerian manufacturing organisations. Similar conclusions were reached by Naz and Khan (2024), who found that fraud investigation, litigation support, and dispute resolution significantly improved fraud detection among firms in Pakistan. Likewise, Kaur et al. (2022) concluded from a systematic review that forensic accounting consistently enhances fraud detection despite implementation challenges. The consistency of these findings across different jurisdictions suggests that forensic accounting has evolved into a globally recognised mechanism for strengthening organisational fraud management.

The findings align with previous studies while highlighting the crucial role of forensic accounting alongside audit quality in fraud detection. Previous research typically assessed forensic accounting in isolation, limiting its comparative influence in governance mechanisms. This study's

stronger coefficient suggests that specialised investigative skills may be more effective in detecting fraud than standard audit processes in manufacturing firms. As financial fraud becomes more sophisticated, it necessitates advanced digital forensic analysis and continuous monitoring.

Moreover, the results support the Fraud Triangle Theory, particularly the opportunity aspect. Enhanced forensic accounting practices can deter fraudulent behaviour by increasing the probability of detection through specialised investigations. Therefore, manufacturing firms should strengthen their forensic units, invest in digital technologies, and develop investigative skills to combat financial crimes, thereby improving transparency and reducing losses from corporate fraud.

The study further demonstrated that audit quality significantly enhances fraud detection, confirming that improvements in auditor independence, competence, professional scepticism, audit planning, and compliance with auditing standards contribute meaningfully to identifying fraudulent financial activities. This finding indicates that high-quality audits remain an essential component of organisational governance, as they strengthen the credibility of financial reporting and increase the likelihood that material irregularities will be detected during audit engagements. Although the influence of audit quality was weaker than that of forensic accounting, its statistical significance demonstrates that effective auditing continues to play a critical role in organisational fraud management.

The result supports previous international evidence linking audit quality with lower incidences of corporate fraud. Nejad et al. (2024) reported that higher audit quality significantly reduced financial statement fraud among Indonesian listed companies, while Almatarneh et al. (2026) found that stronger external audit quality reduced illegal accounting practices within Jordanian firms. Comparable evidence was presented by Bonrath and Eulerich (2024), who observed that effective auditing supported by robust governance structures and technology-based audit processes enhanced fraud prevention and detection. Collectively, these studies reinforce the argument that high-quality auditing strengthens organisational oversight and promotes greater financial accountability.

Notwithstanding these similarities, the present findings differ from Lubis et al. (2024), who found that audit quality did not significantly influence fraud prevention within Indonesian public-sector organisations. Their study attributed fraud prevention primarily to internal audit effectiveness and internal control systems rather than external audit quality. Variations in regulatory environments, organisational structures, governance maturity, and institutional accountability may explain these contrasting findings. Public-sector organisations frequently operate under different accountability frameworks from manufacturing firms, potentially reducing the direct influence of audit quality on fraud detection outcomes.

Beyond confirming previous evidence, this study demonstrates that audit quality should not be viewed solely as a statutory compliance mechanism but as an active contributor to fraud risk management. Professional scepticism, comprehensive audit planning, and rigorous evaluation of audit evidence collectively improve auditors' ability to recognise fraud indicators before financial irregularities escalate. The findings further support Agency Theory by illustrating how independent auditing reduces information asymmetry between organisational managers and owners while

strengthening accountability. In practice, manufacturing firms should continue to invest in auditor competence, independence, and continuous professional development to improve audit effectiveness in increasingly complex business environments.

An important contribution of this study is that forensic accounting and audit quality jointly exert a significant influence on fraud detection, explaining a substantial proportion of the variation in fraud detection among manufacturing firms. Rather than operating as isolated governance mechanisms, the findings indicate that both variables complement one another in strengthening organisational capacity to identify, investigate, and prevent fraudulent activities. The significant overall regression model confirms that integrating specialised investigative procedures with high-quality auditing yields a stronger fraud detection framework than relying exclusively on either mechanism.

This finding addresses an important gap identified in the empirical literature. Previous studies have generally examined forensic accounting and audit quality separately, with limited attention given to their interaction within manufacturing organisations. While earlier studies consistently demonstrated the positive contribution of forensic accounting (Lawal, 2020; David et al., 2026; Naz & Khan, 2024) and others highlighted the importance of audit quality (Nejad et al., 2024; Almatarneh et al., 2026), little empirical evidence had established whether combining both mechanisms produces stronger fraud detection outcomes. The present findings, therefore, extend the existing literature by providing evidence that fraud detection is strengthened when investigative expertise is combined with rigorous audit practices.

The observed relationship between audit quality and forensic accounting enhances fraud detection in manufacturing firms. Audit quality improves fraud risk assessment through systematic planning and objective evaluation, while forensic accounting boosts investigative capacity with digital analysis and litigation support. Their integration creates multiple layers of oversight, reducing opportunities for fraud and increasing the likelihood of detection, particularly in environments with extensive inventories and high transaction volumes. The findings expand on Fraud Triangle Theory by showing how these governance mechanisms collectively mitigate fraud risks. Manufacturing firms should foster collaboration between forensic accountants and auditors, incorporate digital forensic tools into audits, strengthen audit committees, and invest in ongoing professional training. These measures are likely to enhance fraud detection, improve the credibility of financial reporting, and bolster stakeholder confidence.

Conclusion

This study examined the individual and joint effects of forensic accounting and audit quality on fraud detection in manufacturing firms. The findings showed that forensic accounting significantly improves fraud detection by strengthening investigative capacity, enhancing fraud risk assessment, and identifying concealed financial irregularities. Audit quality also exerted a significant positive influence, demonstrating that auditor independence, competence, professional scepticism, and compliance with auditing standards enhance fraud detection. More importantly, the results established

that integrating forensic accounting with high-quality auditing provides a more effective fraud detection framework than relying on either mechanism independently.

The findings indicate that effective fraud management requires an integrated governance approach that combines specialised forensic accounting practices with high-quality auditing. As financial crimes become increasingly sophisticated, conventional audit procedures alone are insufficient to address emerging fraud risks. Incorporating forensic accounting techniques into routine audit processes strengthens organisational capacity to detect, investigate, and prevent fraudulent activities while improving financial reporting credibility and corporate accountability. The study contributes to existing knowledge by providing empirical evidence on the complementary relationship between forensic accounting and audit quality in manufacturing firms, thereby extending the application of Fraud Triangle Theory and Agency Theory to integrated fraud detection.

Recommendations

Based on the findings, manufacturing firms should establish stronger forensic accounting units staffed by qualified specialists and supported with adequate investigative resources. Organisations should reinforce auditor independence through sound governance structures and invest in digital forensic technologies, including data analytics and continuous auditing systems, to improve fraud detection capabilities. Continuous professional training should be provided to auditors and forensic accountants to enhance competence in emerging fraud schemes and investigative techniques. Audit committees should be strengthened to improve oversight of financial reporting and fraud risk management, while forensic accounting procedures should be incorporated into routine audit engagements.

The findings also have important policy implications. Regulatory authorities should strengthen compliance requirements relating to forensic accounting and audit quality, while professional accounting bodies should expand certification and continuing professional development in forensic investigation and digital auditing. Corporate governance frameworks should encourage stronger collaboration among management, auditors, forensic accountants, and audit committees.

The study is limited by its relatively small sample size, cross-sectional design, and reliance on self-reported data, which may restrict generalisability and causal interpretation. Future studies should employ longitudinal designs, compare findings across industrial sectors, examine moderating variables such as corporate governance and organisational culture, and investigate the role of artificial intelligence and digital audit technologies in enhancing forensic accounting and fraud detection.

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